

# Pre-Analysis Plan for Emotions and Costless Credible Signaling Emotional Expressions and Compromise in Bargaining

Sam Sharman\*      Erik Chi<sup>†</sup>      Nathan McCoslin<sup>‡</sup>

May 28, 2026

## Abstract

This experiment falls under a larger project in which we investigate the microfoundations of and the effectiveness of a variety of signals in International Relations across mass and elite samples. In this project we focus on the role of emotional expressions in crisis bargaining, which is beyond the scope of existing rationalist theories of communication. The prevailing paradigm is that only costly signals are informative, but an emerging body of literature focusing on costless cheap talk models of communication challenge this assertion. Existing rationalist models continue to ignore a critical aspect of human communication: emotional expressions. These include numerous non-verbal cues that comprise vital aspects of human interaction and information transmission. This study aims to validate and extend [Yoon \(2026\)](#)'s theory by testing the effectiveness of emotional expressions on compromise in bargaining. We employ an incentivized survey experiment featuring the compromise game, a two-sided version of the ultimatum game.

---

\*Assistant Professor, Graduate School of International Relations, International University of Japan. Email: [ssharman13@gmail.com](mailto:ssharman13@gmail.com). ORCID: 0000-0002-3065-1301.

<sup>†</sup>PhD Candidate, Department of Political Science, Texas A&M University. Email: [erikchi@tamu.edu](mailto:erikchi@tamu.edu).

<sup>‡</sup>PhD Student, Department of Political Science, Texas A&M University. Email: [nathanmccoslin@tamu.edu](mailto:nathanmccoslin@tamu.edu).

# 1 Introduction

What makes a credible threat? A significant part of the conflict literature in International Relations has been dedicated to answering the simple question of how states can credibly signal their intent, resolve, or capabilities to fight (Schelling, 1966). The importance of this question lies in why states may inadvertently go to war with one another when a negotiated peace is both preferable and possible Fearon (1995). In answering this question, scholars have classified signals as either costly or costless (‘cheap talk’) in game theoretic models of state-to-state communication. Although the scholarly consensus regards ‘cheap talk’ messages as uninformative, recent authors have challenged this notion (e.g.: Ramsay, 2004, 2011; Lindsey, 2017; Joseph, 2021; Yoon, 2026). This paper focuses on how costless signals can credibly convey resolve or willingness to fight. It explores one mechanism of costless signaling: emotional expressions.

To assess how emotional expressions during negotiations affect bargaining behavior, we design an incentivized experiment where respondents play multiple rounds of a compromise game. Previous work has examined anger in the one-sided form of these games (ultimatum games), but they focus on the expression of anger (as measured by MRI scans of the brain) in those faced with an unfair offer (Pillutla and Murnighan, 1996; Sanfey et al., 2003). By contrast, we are interested in whether individuals become more compromising when faced with emotional appeals. Furthermore, Yoon (2026) demonstrates the effectiveness of anger expressions in elevating the perceived resolve of opponents in bargaining, but he does not demonstrate changes in behavior.

## 2 Information and War

Fearon (1995)’s seminal contribution to the conflict literature in the *Rationalist Explanations for War* offered three main reasons for why rational states might engage in war despite peacefully negotiated settlements being preferable to the outcomes obtained through war: 1) incentives to misrepresent, 2) commitment problems, and 3) issue indivisibility. Of these three explanations, the first, incentives to misrepresent, offers the most relevant explanation for the question at hand. In the now canonical bargaining model of war, Fearon (1995) shows that states have incentives to over-report their resolve or capabilities to fight in an effort to secure a better bargaining outcome. Both states can freely lie and exaggerate their resolve or capabilities to fight because they do not incur a penalty for doing so. This lack of penalty makes claims of resolve costless (or ‘cheap talk’) and therefore uninformative to the other state; it is always better to lie and overstate one’s capabilities.

Thus, the predominant paradigm among scholars of international conflict posits that signals must be costly to the sender in order to be credible. Indeed, a large body of work has been dedicated to characterizing the mechanisms through which costly, and therefore credible, signals can be sent by one state to another. These include mechanisms such as sunk costs through military mobilization or hand tying through public threats (Fearon, 1997; Fuhrmann and Sechser, 2014). Although the second mechanism may not, *prima facie*, appear to be a costly signal, leaders can incur costs in making public proclamations through audience costs. Domestic constituents may punish the leader for escalating conflicts unnecessarily, escalating

and then backing down, or fighting after promising not to fight (Fearon, 1994; Kertzer and Brutger, 2016; Quek, 2017). A second generation of literature spawning from these game theoretic models has thoroughly tested the microfoundations of audience cost theories using experimental manipulations (Tomz, 2007; Levy et al., 2015; Kertzer and Brutger, 2016). These papers reinforce the idea that credible communication must be conducted through costly means.

Disputing this paradigm that signals must be costly in order to be effective, is a recent body of work focusing on the merits and mechanisms of costless ‘cheap talk’ signaling models. Among these models, Katagiri and Min (2019) argue that a state’s intentions are more easily conveyed through private communications, which are less noisy and more targeted than public ones. Although public signals earn their credibility through the generation of audience costs, their intent is also prone to obfuscation and misattribution. Is the rival conveying resolve, or galvanizing support among a domestic audience? Lindsey (2017) offers another mechanism for costless credible signaling through the role of diplomats. Because ambassadors share an affinity towards their host countries (Halperin, Clapp and Kanter, 2006), disagreements between the ambassador and the president convey important information about the intent and sincerity of the president’s threats. Other still, argue that convergent or divergent signals between a leader and a domestic opposition can provide actionable information about the country’s intentions (Ramsay, 2004). Whereas experimental testing of costly signaling models has been thoroughly conducted, only a meager amount of attention has been paid to these models of costless signaling.

In a recent innovation Yoon (2026) extends the purview of costless signaling mechanisms to expressions of anger during interstate negotiations. Though he does not make the connection explicit, this is another form of costless signaling that relies on the complex and multifaceted non-verbal expressions featured in human communication.

## 3 Information Encoded in Emotion

### 3.1 Anger Expressions

In challenging rationalist accounts that omit the psychological dimensions of communication, Yoon (2026) argues that expressions of anger in interstate negotiations can serve as a source of credibility in cheap talk. His central claim is that anger transmits beliefs and intentions through evolutionary and social mechanisms shared across individuals. This credibility stems from the difficulty of fabricating the nonverbal cues associated with genuine emotional expression.

Although diplomacy is often perceived as a formal and composed process, emotional outbursts are not uncommon (See: Figure 3). This raises an important question: what information do such expressions actually convey?

Anger has well documented behavioral effects. It increases risk-seeking tendencies and heightens individuals’ willingness to incur costs that would otherwise seem excessive (Berkowitz and Harmon-Jones, 2004; Lerner and Keltner, 2001; Frank, 1988). In doing so, it promotes aggression, hostility, and a greater inclination to punish perceived adversaries. In the complex informational environment of interstate negotiations, where multiple signals are often

conveyed simultaneously, leaders may rely on such intuitive and readily interpretable cues (Katagiri and Min, 2019).

Consistent with this logic, anger expressions appear to communicate meaningful information in bargaining contexts. van Leeuwen et al. (2017) show that independent observers can identify individuals who received unfair offers in an ultimatum game based solely on images of their facial expressions. Further, because anger is a universal emotion rooted in shared evolutionary history, its expression is broadly understood across cultures (Sell and Lopez, 2020).

Following this, **Hypothesis 1** states:

**H1:** Respondents exposed to anger expressions during bargaining will be more compromising than the respondents exposed to the placebo control condition.

### 3.2 Appeals to Sympathy

While Yoon (2026) brings attention to expressions of anger and outrage during interstate communications, these emotions are not uniquely represented in negotiations. Indeed, impassioned pleas are common features of interstate negotiations (See: Table 4).

Emotional, often moral, appeals operate through a distinct set of behavioral mechanisms rooted in sympathy and prosocial motivation (Batson and Ahmad, 2001). While rationalist approaches to negotiation suggest that such appeals should be ineffective—or even counterproductive by revealing weakness—experimental evidence indicates that they can systematically increase cooperation and concessions (Shirako, Kilduff and Kray, 2015). Inducing sympathy has been shown to elevate individuals’ willingness to help others and to make more generous offers, even in strategic settings (Batson and Ahmad, 2001). For instance, exposure to another party’s suffering or vulnerability increases prosocial behavior, while perspective-taking enhances cooperation in dilemmas involving conflicting interests (Eisenberg et al., 1989). These effects arise because emotional appeals activate deeply ingrained psychological tendencies toward empathy and moral concern, which in turn shape preferences over outcomes (Haidt, 2003). In bargaining environments characterized by uncertainty and incomplete information, such cues may serve as intuitive signals that the counterpart is deserving of assistance or consideration, thereby increasing the likelihood of voluntary concessions.

Therefore, **Hypothesis 2** states:

**H2:** Respondents exposed to the emotional plea during bargaining will be more compromising than the respondents exposed to the placebo control condition.

However, given that bargaining is an action frequently placed in the context of zero sum interactions, we should expect that anger expressions induce respondents to be more compromising relative to the emotional plea treatment. This is because anger signals greater risk-seeking and greater proclivity towards reckless self-destructive behavior. In the context of zero-sum bargaining, we may expect that anger expressions signal a willingness to tank the negotiations all together. Conversely, emotional pleas may generate sympathies, but these expressions do not convey a willingness to deviate from rationalist calculations and leave both parties with nothing.

**Hypothesis 3** therefore states:

**H3:** Respondents exposed to anger expressions during bargaining will be more compromising than the respondents exposed to the emotional plea condition.

## 4 Design

We propose an incentivized experiment with a repeated measures design to estimate the inducement of compromise that each type of emotional appeal may or may not induce within a respondent. Respondents will play multiple rounds of a two-sided version of the ultimatum game known as the compromise game. In contrast to the ultimatum game, which only permits a single interaction in the form of a *fait accompli*, the compromise game features symmetric bargaining power and helps players reach more equitable outcomes.

[Delton, DeScioli and Ryan \(2020\)](#) employ a similar strategy to study moral obstinacy in political negotiation, but they only permit one round of interactions between a respondent and a given opponent. This is appropriate given the quantity of interest in their study, which is the proportion of successful bargains by moral obstinacy and by political context. By contrast, we are interested in the degree of compromise induced by emotional outbursts within an individual with respect to the same opponent across two rounds of bargaining.

We deviate further from [Delton, DeScioli and Ryan \(2020\)](#) by having all respondents play the compromise game against a bot with prespecified moves. Since the experimental conditions (explained below) all consist of pre-recorded messages from actors, there is no advantage to having respondents play against each other. Instead, all respondents face a demand from the bot requesting an additional unit from the respondent. If the division of the monetary incentive ranges from 0% to 100% then the possible set of divisions is represented by an ordered pair whose elements sum to 100%. The first entry is the proportion retained by the respondent and the second entry is the proportion offered to the other player (the bot). (E.g.: (respondent, bot).)

### 4.1 Sequence of Play

Respondents are introduced to the compromise game and the monetary stakes of the game are explained to the respondents. The game is played in two iterations with two rounds each.

The first iteration presents the same content to all respondents so that we can measure the pre-treatment level of compromise for each respondent. This allows us to incorporate the pre-treatment measure of the outcome as a control variable and improve statistical precision. While there are concerns with demand effects or consistency pressures, evidence suggests that the biases induced by these sources are negligible ([Clifford, Sheagley and Piston, 2021](#)).

The experiment plays out in the following sequence. First the respondent makes an initial offer (call this  $(r_{11}, b_{11})$ ). The bot always plays off of the respondent's initial offer and requests  $(r_{11} - 10\%, b_{11} + 10\%)$  so that there is always a demand for one additional incremental unit. After making the initial offer and viewing the bot's counter offer, the respondents are shown a video. This is a pre-recorded message from an actor displaying a calm demeanor. The actor asks that the respondent acquiesces to the demand of one additional unit. This

functions as the placebo video stimulus. The respondent then makes the second round offer  $(r_{12}, b_{12})$  and the bot maintains their first round offer of  $(r_{11} - 10\%, b_{11} + 10\%)$ .

We then take the respondents to a second iteration of the game with another two rounds of bargaining. This iteration is separate from the first and the respondents are informed that there is a new player on the other end. We then randomize assignment into the control, anger, or emotional plea conditions for the second iteration. Another two rounds are played with the the respondent's offer of  $(r_{21}, b_{21})$  and the bot's counter of  $(r_{21} - 10\%, b_{21} + 10\%)$ . Respondents view the manipulation associated with their assigned experimental condition and make their second round offer of  $(r_{22}, b_{22})$ . Again, the bot's response is held fixed at their first round offer of  $(r_{21} - 10\%, b_{21} + 10\%)$ . For an illustrative diagram of the experiment's flow see Figure 1 in the appendix.

There is one additional complication that arises from the possibility of overly generous first round offers. To illustrate, suppose the respondent's first round offer is  $(0\%, 100\%)$ , then the bot can not request an additional unit because the proposition must sum to 100%. These types of respondents would simply skip the second round of bargaining. Depending on the measurement of compromise, the existence of these types of respondents can be dealt with in different ways. We discuss the measure of compromise in the next section before addressing these concerns.

## 4.2 Quantity of Interest

We measure compromise in two ways. First, we measure whether or not the respondent came to a compromise in the second round within a given iteration of the compromise game as a binary variable. For example, if the respondent made a offer  $(50\%, 50\%)$  in the first round and failed to offer an additional unit in the second round, then compromise was not reached. Consequently, the compromise variable would be coded as a zero for this respondent. If, however, their second round offer was  $(40\%, 60\%)$ , then compromise was reached and the variable would be coded as a one.

A second way that we can measure compromise is by the degree of compromise or the difference in offers between the first and the second rounds. Suppose the respondent's first round offer was  $(60\%, 40\%)$  and their second round offer was  $(40\%, 60\%)$ , then the respondent compromised by twenty percentage points. This version of the measurement may give us higher variation in the outcome as compared to the dichotomized measure of compromise. From the hypotheses we would expect that the likelihood or degree of compromise among respondents in the treatment groups to be larger than the degree of compromise among the respondents in the control group.

Recall, the potential issue raised by the existence of overly generous first round offers. This issue arises because the bot can not request more than the maximal division of 100%. With respect to the difference measure, there is not much that can be done. We may choose to drop these respondents from the sample of analysis. However, the dichotomized measure offers an alternative that may prevent this data loss. To deal with these types of respondents in the dichotomous measure, we can simply classify these respondents as having compromised. If, however, the respondent continues to be extremely generous in the second iteration of the game, then they would not be eligible to enter the sample of analysis because they would skip over the treatment.

Additional details on the outcome measures in the appendix.

### 4.3 Resulting Sample of Analysis

Following the discussion of extremely generous respondents above there are a number of complications in constructing the sample of analysis that varies from one measure to another. The difference measure is incapable of dealing with the extremely generous respondents because they skip the second round of bargaining within a given iteration; no difference can be taken to measure the degree of compromise. The resulting sample of analysis would filter out these overly generous respondents.

The binary measure of compromise, however, can accommodate some of the extremely generous respondents, just not all of them. Respondents who are extremely generous in the first round of the first iteration can have their pre-treatment measure of compromise equal to one (as opposed to zero). However, those who continue to be extremely generous in the first round of the second iteration can not enter the sample of analysis because they have maximally compromised prior to treatment exposure. Therefore, the sample of analysis using the binary measure of compromise would exclude the respondents who were maximally generous in the first rounds of both iterations.

Additional details for the inclusion or exclusion of respondents are in the appendix.

### 4.4 Experimental Conditions

Respondents are randomly assigned to one of two treatment conditions: 1) Anger and 2) Plea. In the anger condition, the actor angrily expresses their demand for a larger allocation, where as in the plea condition they ask for a larger share with an emotional appeal about why they need the larger share. Respondents assigned to the control condition are provided a video in which the actor asks for a greater share with a calm demeanor. This is essentially the same type of video shown to the respondents in the first iteration of the game.

The plan is to have multiple actors record videos for each condition, so that reviewers do not argue that our results are solely driven by the features present in one particular actor's performance. A diversity of faces might also help make the situation more believable for the respondent.

Transcriptions of each type of video treatment are listed in the appendix.

### 4.5 Realism of the Experiment

If respondents can not send their own video message, then they are more likely to identify the treatment as the video. Consequently, some respondents will not take the video or the remainder of the survey very seriously. This could potentially compromise the experiment. To avoid this issue the introduction to the survey experiment will employ deception in the framing of the experiment. We will frame the survey experiment as a study on one-sided communication. We will tell the respondents that some of them will be selected to only receive the video message without the option to send their own. This way we can still present the videos as messages from another player. However, in reality, none of the respondents will be able to send messages of their own and they are playing against a bot. Ideally, this

strategy helps the respondents suspend their disbelief and rationalize why they are not able to send their own messages.

## 5 Analysis Strategy

From the experimental design we have two measures of compromise. One constructed as a binary measure of whether or not the respondent compromised in the second round, and another as the difference between the first and second round offers (i.e. how much they compromised by). As was discussed, the two different measures allows for the inclusion, or exclusion, of different parts of the sample. To reiterate: the binary measure allows for the inclusion of a larger proportion of the sample into the analysis because it can incorporate the respondents who propose maximal offers in the first round.

Irrespective of the outcome measure used, the analysis is simple. The unit of analysis is the respondent, and we regress the measure of compromise in the second iteration of the compromise game on a dummy variable equal to one if the respondent belongs to the anger condition, a dummy variable equal to one if the respondent belongs to the emotional plea condition, and the measure of compromise from the first iteration of the compromise game. This is given in Equation 1 as follows:

$$y_i^{post} = \beta_0 + \beta_1 \text{Anger}_i + \beta_2 \text{Plea}_i + \beta_3 y_i^{pre} + \epsilon_i \quad (1)$$

Here  $y_i^{post}$  denotes the measure of compromise from the second iteration of the game,  $y_i^{pre}$  the compromise measure from the first iteration,  $\text{Anger}_i$  a dummy variable equal to one if the respondent belongs to the Anger condition,  $\text{Plea}_i$  a dummy variable equal to one if the respondent belongs to the Emotional Plea condition, and  $\epsilon_i$  the idiosyncratic error term.

Substantively, the intercept  $\beta_0$  captures the average level of second iteration compromise within the control group, though we have no *ex ante* theory driven expectations about this coefficient.

As for  $\beta_1$  we expect the coefficient, as per **Hypothesis 1**, to be positive. The substantive interpretation derived from this is that respondents assigned to the Anger condition are more likely (in the binary measure) or exhibit a greater degree of compromise (in the difference measure).

Following **Hypothesis 2** also we expect  $\beta_2$  to be positive. This would indicate that the respondents assigned to the Emotional Plea condition were more likely or exhibited a greater degree of compromise relative to the control group.

For the test of **Hypothesis 3** we compare the magnitude of  $\beta_1$  and  $\beta_2$ . If both are positive following Hypotheses 1 and 2, then Hypothesis 3 predicts that  $\beta_2 > \beta_1$ . We test this with the linear hypothesis  $\beta_2 - \beta_1 > 0$ .

The inclusion of  $y_i^{pre}$  follows from the repeated measures design of this experiment and it's associated coefficient is not to be interpreted. It's inclusion improves the precision of our estimates by reducing residual variation.

For the hypothesis tests specified above, we set the size of the test equal to 5% or  $\alpha = 0.05$ . The necessity of post-estimation standard error corrections can be tested using the Breusch-

Pagan or the White test. If the null of these tests can be rejected (also at the 5% level), then we can employ heteroskedasticity robust standard errors.

## References

- Batson, C. Daniel and Nadia Ahmad. 2001. "Empathy-Induced Altruism in a Prisoner's Dilemma II: What If the Target of Empathy Has Defected?" *European Journal of Social Psychology* 31(1):25–36.
- Berkowitz, Leonard and Eddie Harmon-Jones. 2004. "Toward an Understanding of the Determinants of Anger." *Emotion* 4(2):107–130.
- Clifford, Scott, Geoffrey Sheagley and Spencer Piston. 2021. "Increasing Precision Without Altering Treatment Effects: Repeated Measures Designs in Survey Experiments." *American Political Science Review* 115(3):1048–1065.
- Delton, Andrew W., Peter DeScioli and Timothy J. Ryan. 2020. "Moral Obstinacy in Political Negotiations." *Political Psychology* 41(1):3–20.
- Eisenberg, Nancy, Richard A. Fabes, Paul A. Miller, Jim Fultz, Rita Shell, Robin M. Mathy and Ray R. Reno. 1989. "Relation of Sympathy and Personal Distress to Prosocial Behavior: A Multimethod Study." *Journal of Personality & Social Psychology* 57(1):55.
- Fearon, James D. 1994. "Domestic Political Audiences and the Escalation of International Disputes." *American Political Science Review* 88(3):577–592.
- Fearon, James D. 1995. "Rationalist Explanations for War." *International Organization* 49(3):379–414.
- Fearon, James D. 1997. "Signaling Foreign Policy Interests: Tying Hands Versus Sinking Costs." *Journal of Conflict Resolution* 41(1):68–90.
- Frank, Robert H. 1988. *Passions Within Reason: The Strategic Role of the Emotions*. W.W. Norton.
- Fuhrmann, Matthew and Todd S. Sechser. 2014. "Signaling Alliance Commitments: Hand-Tying and Sunk Costs in Extended Nuclear Deterrence." *American Journal of Political Science* 58(4):919–935.
- Haidt, Jonathan. 2003. The Moral Emotions. In *Handbook of Affective Sciences*, ed. Richard J. Davidson, Klaus R. Scherer and H. Hill Goldsmith. Oxford University Press.
- Halperin, Morton H., Priscilla A. Clapp and Arnold Kanter. 2006. *Bureaucratic Politics and Foreign Policy*. 2nd ed. Brookings Institution Press.
- Joseph, Michael F. 2021. "A Little Bit of Cheap Talk Is a Dangerous Thing: States Can Communicate Intentions Persuasively and Raise the Risk of War." *The Journal of Politics* 83(1):166–181.
- Katagiri, Azusa and Eric Min. 2019. "The Credibility of Public and Private Signals: A Document-Based Approach." *American Political Science Review* 113(1):156–172.

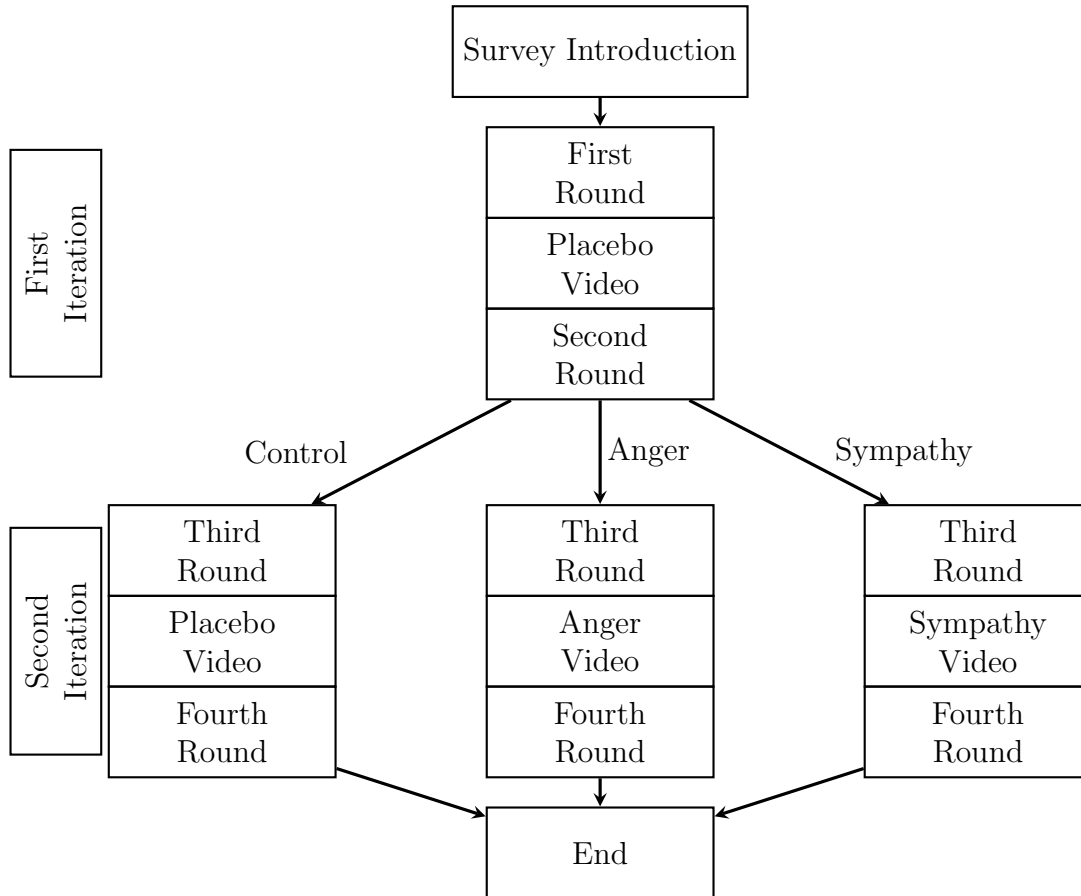
- Kertzer, Joshua D. and Ryan Brutger. 2016. "Decomposing Audience Costs: Bringing the Audience Back Into Audience Cost Theory." *American Journal of Political Science* 60(1):234–249.
- Lerner, Jennifer S. and Dacher Keltner. 2001. "Fear, Anger, and Risk." *Journal of Personality & Social Psychology* 81(1):146.
- Levy, Jack S., Michael K. McKoy, Paul Poast and Geoffrey P.R. Wallace. 2015. "Backing Out or Backing In? Commitment and Consistency in Audience Costs Theory." *American Journal of Political Science* 59(4):988–1001.
- Lindsey, David. 2017. "Diplomacy Through Agents." *International Studies Quarterly* 61(1):544–556.
- Pillutla, Madan M. and J. Keith Murnighan. 1996. "Unfairness Anger, and Spite: Emotional Rejections of Ultimatum Offers." *Organizational Behavior and Human Decision Processes* 68(3):208–224.
- Quek, Kai. 2017. "Type II Audience Costs." *The Journal of Politics* 79(4):1438–1443.
- Rainey, Carlisle. 2026. "Power Rules: Practical Advice for Computing Power (and Automating With Pilot Data)."   
**URL:** [osf.io/5am9qv3](https://osf.io/5am9qv3)
- Ramsay, Kristopher W. 2004. "Politics at Water's Edge: Crisis Bargaining and Electoral Competition." *Journal of Conflict Resolution* 48(4):459–486.
- Ramsay, Kristopher W. 2011. "Cheap Talk Diplomacy, Voluntary Negotiations, and Variable Bargaining Power." *International Studies Quarterly* 55(4):1003–1023.
- Sanfey, Alan G., James K. Rilling, Jessica A. Aronson, Leigh E. Nystrom and Jonathan D. Cohen. 2003. "The Neural Basis of Economic Decision-Making in the Ultimatum Game." *Science* 300(5626):1755–1758.
- Schelling, Thomas C. 1966. *Arms and Influence*. Yale University Press.
- Sell, Aaron N. and Anthony C. Lopez. 2020. Emotional Underpinnings of War: An Evolutionary Analysis of Anger and Hatred. In *The Handbook of Collective Violence: Current Developments and Understanding*, ed. Carol A. Ireland, Michael Lewis, Anthony Lopez and Jane L. Ireland. Routledge.
- Shirako, Aiwa, Gavin J. Kilduff and Laura J. Kray. 2015. "Is There Place for Sympathy in Negotiation? Finding Strength in Weakness." *Organizational Behavior & Human Decision Processes* 131:95–109.
- Tomz, Michael. 2007. "Domestic Audience Costs in International Relations: An Experimental Approach." *International Organization* 61(4):821–840.

- van Leeuwen, Boris, Charles N. Noussair, Theo Offerman, Sigrid Suetens, Matthis van Veelen and Jeroen van de Ven. 2017. "Predictably Angry – Facial Cues Provide a Credible Signal of Destructive Behavior." *Management Science* 64(7):2973–3468.
- Yoon, Hohyun. 2026. "Anger Expression and Coercive Credibility in International Crises." *American Journal of Political Science* 70(1):221–237.

# A1

## Survey Diagram

Figure 1: Flow of the experiment.



We employ a number of actors to record each type of message. The videos are played prior to the second round of each iteration of the compromise game.

The above section demonstrates the flow of the experiment. The following sections illustrate the details provided to respondents.

## Survey Introduction

We are conducting an experiment on one-sided communication and how it influences respondents in the compromise game as compared to two-sided communication.

In the compromise game you will be eligible to earn a monetary reward up to \$1 dollars through your actions against the other respondent in addition to the \$1 dollar you are earning by taking this survey.

You will play the compromise game against two other respondents who are taking the same survey. There will be two opportunities to bargain against each of the respondents you are playing against. In total, there are four rounds of bargaining and two opportunities to earn the additional monetary reward.

Against each opponent the game will be played through the same three phases. First, both players will offer a division of the monetary reward between 0% and 100% in increments of 10% to their opponent. If the two offers are incompatible, then the round of bargaining fails and no monetary reward is paid out. Second, if bargaining has failed, then one or both players may be selected to send a message to the other. Third, another opportunity is provided to bargain with the same opponent. You will again make offers 0% and 100% in increments of 10% to each other. If a deal is not struck here, then the monetary incentive is lost.

Following these two rounds of bargaining, you will play the game again with a new player.

## Attention Check

Serving as an attention check and helping to ensure that the respondent understands the nature of the game, the respondent will answer a multiple choice question about the compromise game:

We consider respondents attentive if they score  $\frac{3}{4}$  of the following questions correctly. We exclude respondents from the analysis if they fail the attention check. Order is randomized.

1. Q: How much money can you earn through your actions in the bargaining games?
  - \$1.\*
  - \$2.
  - \$3
  - \$0.50
2. Q: How many opportunities are there to bargain against each opponent?
  - 2 per player.\*
  - 3 per player.
  - 4 per player.

- Just once per player.
3. Q: What is the largest proportion of the monetary incentive can you offer to the opposing player?
- 100%.\*
  - 50%
  - 75%
  - 90%
4. Q: If a mutually compatible deal is not made with the other player, will you receive the extra monetary reward for that round?
- No.\*
  - Yes.

## Offers

Respondents designate the proposed division of the monetary incentive by proposing ordered pairs of percentages corresponding to what they would like to keep and what they are offering the other player. Instead of having the respondents input each offer, the following selection is given to:

Please select your offer for this round of bargaining. The first percentage in the pair corresponds with the share you would like to keep and the second percentage in the pair corresponds with the share you are offering to the other player.

|               |            |            |            |            |     |            |            |            |            |
|---------------|------------|------------|------------|------------|-----|------------|------------|------------|------------|
| (Keep, Offer) | (0%, 100%) | (10%, 90%) | (20%, 80%) | (30%, 70%) | ... | (70%, 30%) | (80%, 20%) | (90%, 10%) | (100%, 0%) |
|---------------|------------|------------|------------|------------|-----|------------|------------|------------|------------|

Table 1: Boxes will be resized, but this how respondents choose their proposals

## Stimulus

Placebo/control and treatment videos are commissioned through actors on the popular free-lance platform Fiverr. We hire three or four actors per condition. The actor is instructed to record the following transcripts:

## Outcome Measurement

As discussed in the main text there are two ways to measure the outcome of compromise. One is a dichotomous measure and another is a difference measure. Depending on the measure, more or less of the sample is eligible to enter the analysis. The details of inclusion/exclusion from the sample of analysis are discussed above.

| Condition                 | Transcript                                                                                                                                                                                                                                                                         |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Placebo/Control</b>    | [Calm demeanor.] “I’d like to ask you to reconsider your previous offer. I think I deserve a better offer than that. For this next round, I’m asking you to make a more generous offer. I hope you’ll reconsider in the next round.”                                               |
| <b>Anger Expression</b>   | [Forceful and outraged.] “I can’t believe you’re being so ridiculous. I was already screwed over by the last guy. This time, you need to make a more generous offer. Don’t send another insulting proposal.”                                                                       |
| <b>Appeal to Sympathy</b> | [Anxious.] “I was really disappointed by your first offer and I really need this. Please, in this second round, I’m asking you to make a more generous offer. I hope you’ll think about how it would feel to be in my position and what would be fair if our roles were reversed.” |

Table 2: Example transcripts for the video stimulus.

The dichotomous measure is equal to 1 if the respondent’s first round offer is 100%, or if the respondent successfully compromises in the second round by offering at least 10% more than their first round offer. (The bot always requests 10% more than the respondent offers unless the first round offer is 100%.)

The difference measure is equal to the difference in the first and second round offers. So, if the first round offer is (80%, 20%) and the second round offer is (60%, 40%), then the degree of compromise is by 20 percentage points.

## Power Analysis

Following [Rainey \(2026\)](#) I calculate the necessary sample size (per condition) for 80% power for an effect size of 0.05, or 5% more likely to reach a bargain. The following figures are reported for the binary measure of compromise rather than the difference measure.

Employing [Rainey \(2026\)](#)’s Rule 6 requires that there is a reference population from which a standard deviation of the outcome can be referred to as a baseline. However, existing surveys using the compromise game are relatively uncommon. Therefore, to obtain a best guess for  $SD(Y)$  I refer to [Delton, DeScioli and Ryan \(2020\)](#)’s study on moral obstinacy, which also uses the compromise game. In particular, I take the standard deviation of their outcome (successful compromise) from their nonpolitical compromise game featured in Study 2 as follows:

$$E[\tilde{Y}] = 0.84 \tag{2}$$

$$V[\tilde{Y}] = 0.1344 \tag{3}$$

$$SD[\tilde{Y}] = 0.3666 \tag{4}$$

Assuming that the repeated measures design featured in this study allows us to assume an  $\tilde{R}^2$  of 40%, we can use the following formula to obtain the number of respondents per condition necessary for 80% power:

$$n = 2 \times \left( \frac{2.5 \times SD(\tilde{Y}) \times \sqrt{(1 - \tilde{R}^2)}}{\tilde{\tau}} \right)^2. \quad (5)$$

Plugging everything in we have:

$$n = 2 \times \left( \frac{2.5 \times 0.3666 \times \sqrt{(1 - 0.40)}}{0.05} \right)^2, \quad (6)$$

$$n = 403. \quad (7)$$

Or 403 respondents per condition.

However, this estimated sample size is obtained by using information from a convenience sample of relatively professionalized respondents from [Delton, DeScioli and Ryan \(2020\)](#)'s MTurk survey. This may have had the effect of increasing the mean compromise rate and decreasing the variance and standard deviation of the outcome.

For a more realistic estimate of the necessary sample size we can recalculate the sample size with a larger standard deviation.

Suppose we have:

$$E[\tilde{Y}] = 0.5 \quad (8)$$

$$V[\tilde{Y}] = 0.25 \quad (9)$$

$$SD[\tilde{Y}] = 0.5. \quad (10)$$

Then,

$$n = 2 \times \left( \frac{2.5 \times 0.5 \times \sqrt{(1 - 0.40)}}{0.05} \right)^2, \quad (11)$$

$$n = 750. \quad (12)$$

And the number of respondents per condition needed for 80% power increases to 750. We will aim to target a middle ground with a total sample size of 1500.

## Missing Observations

We exclude observations from the sample of analysis if they did not complete the survey.

## A2

### Anger Examples

Table 3: Examples of Anger Expressions (See: [Yoon \(2026\)](#)).

| Speaker                 | Date          | Target     | Quote                                                                                                                                                       |
|-------------------------|---------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jimmy Carter            | Jan. 13, 1980 | Iran       | “We are outraged that, half a world away, the Iranian Government holds 50 innocent Americans hostage.”                                                      |
| Josip Tito              | Oct. 11, 1953 | Italy      | “It is not only our people who will not forget it and who are indignant at it, but I already know that abroad also there is indignation at such an action.” |
| Anastasio Somoza García | Dec. 29, 1978 | Costa Rica | “We are losing our patience. They have kidnapped and taken many of our people and have not returned them to us.”                                            |
| Sukarno                 | Mar. 14, 1964 | Malaysia   | “He said that Indonesia is a madman. . . We are called madmen, madmen, madmen.”                                                                             |

## Plea Examples

Table 4: Examples of Pleas.

| Speaker               | Date           | Target    | Quote                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------|----------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Franklin D. Roosevelt | Apr. 14, 1939  | Germany   | “I think you will not misunderstand the spirit of frankness in which I send you this message. Heads of great governments in this hour are literally responsible for the fate of humanity in the coming years. They cannot fail to hear the prayers of their people to be protected from the foreseeable chaos of war. History will hold them accountable for the lives and the happiness of all—even unto the least. I hope that your answer will make it possible for humanity to lose fear and regain security for many years to come.” |
| Volodymyr Zelenskiy   | May 26, 2024   | World     | “I am appealing to the leaders of the world who are still aside from the global efforts of the Global Peace Summit – to President Biden, the leader of the United States, and to President Xi, the leader of China.”                                                                                                                                                                                                                                                                                                                      |
| Yitzhak Rabin         | Sept. 13, 1993 | Palestine | “We who have fought against you, the Palestinians, we say to you today in a loud and a clear voice: Enough of blood and tears. Enough!”                                                                                                                                                                                                                                                                                                                                                                                                   |
| Haile Selassie        | May 5, 1941    | World     | “It is us today. It will be you tomorrow. . . . It is international morality that is at stake.”                                                                                                                                                                                                                                                                                                                                                                                                                                           |